

REQUEST FOR HARDWARE SUPPORT

Hardware components and technical support for SecureMesh, a hardware-enabled decentralised communication system

TO

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Subject: Request for hardware components and technical support for **SecureMesh** — Smart India Hackathon 2026.

Respected Sir/Madam,

We, the undersigned students of the Department of Artificial Intelligence & Machine Learning, are participating in **Smart India Hackathon 2026** with our project **SecureMesh** — a hardware-enabled decentralised communication and situational-awareness system for disaster and network-denied environments.

THE PROBLEM WE ARE ADDRESSING

During natural disasters, remote rescue operations and other emergencies, conventional communication infrastructure such as cellular networks and the Internet may become unavailable. Field teams are then unable to communicate, share their positions, report incidents or maintain a common picture of the situation — precisely when those capabilities matter most.

SecureMesh addresses this by forming a **self-organising, secure peer-to-peer network** that operates with no Internet connection, no cellular network and no central server. A node that can see no peers is in a normal state, not a failed one.

WHAT IS ALREADY WORKING, AND WHAT IS NOT

We have completed a functioning software prototype. We consider it important to state plainly which capabilities are built and demonstrable today and which remain design work, so that the committee can judge this request accurately.

IMPLEMENTED AND DEMONSTRABLE	DESIGNED, NOT YET BUILT
• Cryptographic node identity (Ed25519) • Append-only signed event log • Server-free peer discovery on a local network • Authenticated, encrypted QUIC transport • Operator-enrolled peer trust — authentication alone grants nothing • Incident reporting and conflict-free replication across partitions and restarts	• Long-range LoRa radio link • Dedicated GNSS receiver input • Hardware-protected application signing key • On-device AI inference and retrieval • Battery, enclosure and custom PCB

• Location capture with honest provenance labelling • Periodic location heartbeat between enrolled nodes • Fully offline vector basemap	
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The requested components correspond to the second column. Our objective for this phase is to convert the software prototype into a **physical working node**.

PROPOSED FIRST PROTOTYPE

To keep the initial outlay low, we propose building **one** physical SecureMesh node on a **Raspberry Pi 5 (8 GB)**. Our existing laptop will serve as the second node and a mobile phone as the display, so that a two-node mesh can be demonstrated without purchasing a second board.

The demonstration will show the physical node and the laptop discovering one another, authenticating, exchanging incidents and location reports, and continuing to operate with all Internet connectivity physically disconnected.

We have separately evaluated the NVIDIA Jetson Orin Nano, Orin NX and NXP i.MX 8M Plus as platforms for the later AI phase. The Raspberry Pi 5 is proposed here deliberately instead: this phase carries **no AI workload**, so the accelerator those boards provide would go unused, and the Pi 5 offers the GPIO, SPI and I²C interfaces the radio, GNSS and security modules require at a fraction of the cost.

REQUIRED COMPONENTS

SL.	COMPONENT	QTY	APPROX. COST (₹)
1	Raspberry Pi 5 — 8 GB Single-board computer hosting the node	1	16,000 – 18,000
2	Raspberry Pi 5 active cooler Sustained load without thermal throttling	1	400 – 700
3	128/256 GB high-endurance microSD card Continuous append-only log writes	1	600 – 1,200
4	SX1262 LoRa communication module Long-range link beyond Wi-Fi range	1	700 – 1,200
5	Compatible LoRa antenna	1	250 – 400
6	u-blox NEO-M8N GNSS module Satellite-derived position, independent of any network	1	700 – 1,300
7	GNSS antenna	1	150 – 400
8	TPM 2.0 / hardware security module SPI module compatible with the Pi 5 header — see note below	1	2,000 – 3,500
9	5 V / 5 A USB-C power supply Required for full peripheral current on the Pi 5	1	800 – 1,200
10	USB-C cable	1	200 – 400
11	Breadboard / prototyping board	1	150 – 300
12	Jumper wires and connectors	1 set	300 – 500
	Estimated total		22,000 – 26,000

Costs are indicative retail estimates gathered for planning and are subject to confirmation against supplier quotation at the time of purchase.

We request sanction of approximately **₹30,000** to accommodate price variation, shipping charges and minor consumables required during assembly.

A NOTE ON THE SECURITY MODULE

WHAT ITEM 8 PROTECTS, STATED PRECISELY

We wish to be exact about this component rather than describe it as general “hardware-backed security”. Our platform study established, from the source of the libp2p-identity 0.2.14 library on which our transport depends, that the QUIC transport identity **cannot** be held in a non-exportable hardware key: the keypair type is closed to extension and its signing path requires the private key to be present in process memory.

A TPM therefore cannot protect the transport key, and we do not claim that it will. What it *can* protect is the **application signing key** — the key that signs event-log entries, message envelopes and peer-trust decisions, all of which our own code signs directly. The module is requested so that we can build and measure this separation on real hardware and report honestly on what it does and does not secure.

SUBSEQUENT DEVELOPMENT

The first prototype targets core communication and situational awareness only. Local AI and retrieval functionality are **not** part of this Raspberry Pi implementation.

Following successful validation we intend to extend the system to multiple physical nodes and, at a later stage, introduce an **NVIDIA Jetson Orin Nano** for on-device inference and retrieval. We also intend to improve the physical design with a rechargeable battery system, a rugged enclosure, an improved antenna arrangement and custom PCB integration.

OUR REQUEST

We kindly request the support of the departments in providing or procuring the components listed above, and — where possible — access to laboratory facilities and technical guidance for hardware assembly, testing and integration.

This hardware will allow us to move from a software prototype to a functional physical prototype, which will materially strengthen our demonstration and evaluation at Smart India Hackathon 2026. We assure you that the components will be used exclusively for the development, testing and demonstration of the SecureMesh project, and will be returned to the department where required.

We would be grateful for your consideration and support.

Thank you.

Yours faithfully,

Team SecureMesh

Smart India Hackathon 2026 — Problem Statement STH260001

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ATTACHMENTS

1. SecureMesh project brief
2. Hardware component requirement and estimated cost
3. SIH 2026 registration and problem statement details
4. Platform evaluation — security and compute platform study